

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file <FirstLast>_A07_GLMs.Rmd (replacing <FirstLast> with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

#1

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr    1.5.1
```

```
## v ggplot2    3.5.1      v tibble     3.2.1
```

```
## v lubridate  1.9.4      v tidyr      1.3.1
```

```
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()    masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
```

```
library(lubridate)
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025

getwd()

## [1] "/home/guest/EDE_Fall2025"

here()

## [1] "/home/guest/EDE_Fall2025"

NTLLTER_raw <- read.csv(here("./Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
                        stringsAsFactors = TRUE)

#2
Liz_theme <- theme_grey (base_size = 10) +
  theme(
    plot.title = element_text(
      size = 14,
      face = "bold",
      hjust = 0.5,
      margin = margin(b=10), color = "#FAF0E6"),
    axis.title.x = element_text(color = "#FAF0E6", size = 10, face = "bold"),
    axis.title.y = element_text(color = "#FAF0E6", size = 10, face = "bold"),
    axis.text.x = element_text(color = "#FAF0E6"),
    axis.text.y = element_text(color = "#FAF0E6"),
    plot.background = element_rect(fill = "#008080"),
    panel.background = element_rect(fill = "#FAF0E6"),
    legend.position = "right",
    legend.title = element_text(
      face = "bold"
    )
  )

theme_set(Liz_theme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: There is no change in lake temperature recorded during July with depth across all lakes Ha: There is a change in lake temperature recorded during July with depth across all lakes
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
NTLLTER_raw$sampleddate <-ymd(NTLLTER_raw$sampleddate)
```

```
## Warning: 33138 failed to parse.
```

```
class(NTLLTER_raw$sampleddate)
```

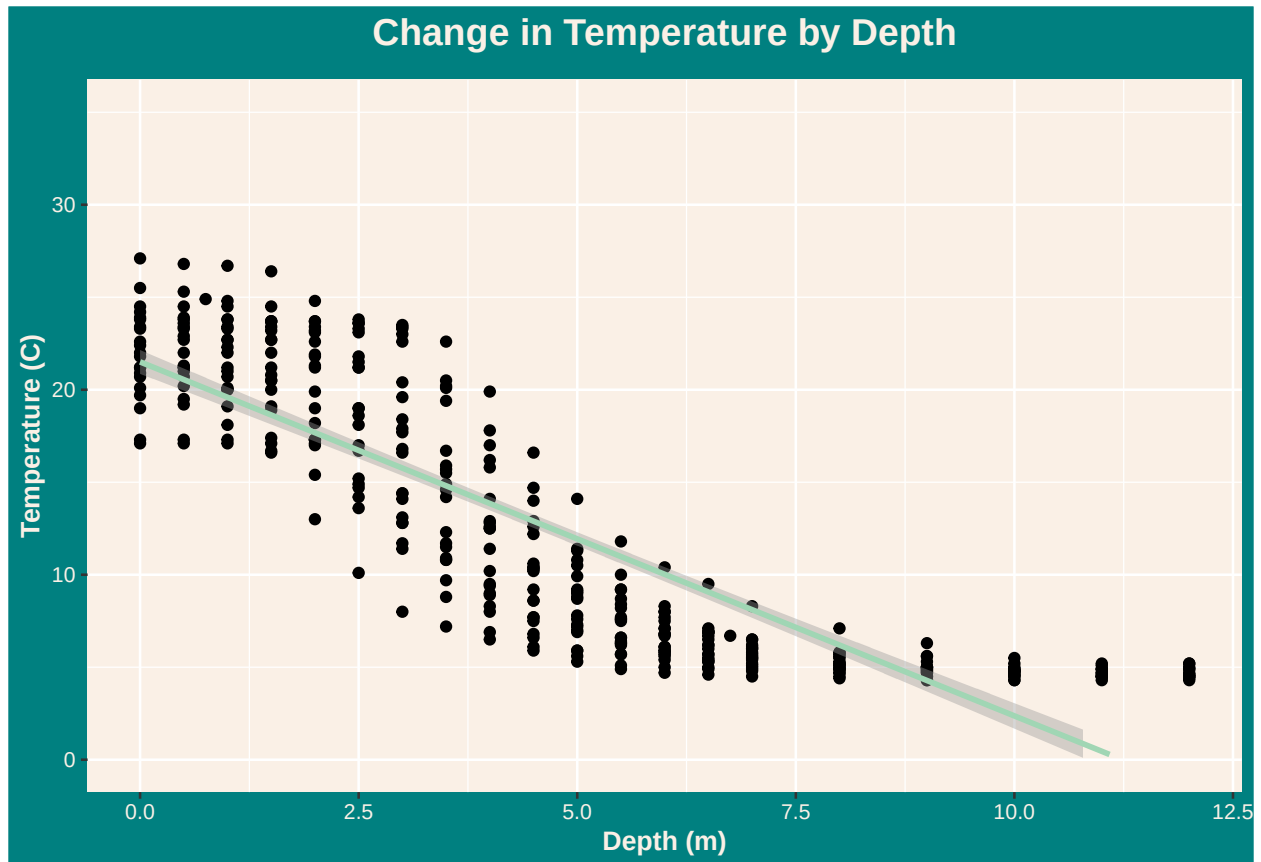
```
## [1] "Date"
```

```
NTLLTER_processed <- NTLLTER_raw %>%
  mutate(month = month(sampledate)) %>%
  filter(month == "7") %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  drop_na()
```

```
#5
Temp.depth <- ggplot(NTLLTER_processed, aes(x=depth, y=temperature_C,)) +
  ylim(0, 35) +
  geom_point() +
  geom_smooth(method="lm", color = "#A0D6B4") +
  labs (title = "Change in Temperature by Depth",
        x = "Depth (m)",
        y = "Temperature (C)")
print(Temp.depth)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: There is a negative correlation and inverse relationship between Temperature and Depth. As Depth increases, the temperature decreases. The points show an s-shaped curve and suggest logistic growth trend.

7. Perform a linear regression to test the relationship and display the results.

```
#7
Temp.depth.regression <- lm(data = NTLLTER_processed, temperature_C ~ depth)
summary(Temp.depth.regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = NTLLTER_processed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.7641 -2.8586 -0.3779  2.6155  7.7928
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   21.5054     0.3261   65.95  <2e-16 ***
```

```
## depth          -1.9138      0.0567  -33.76   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.65 on 392 degrees of freedom
## Multiple R-squared:  0.744, Adjusted R-squared:  0.7434
## F-statistic: 1139 on 1 and 392 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The degrees of freedom is 392 which represents a large sample size that helps increase our confidence in the data. Our intercept is -1.9138, which means that for every meter increase in depth, we can expect a decrease in 1.91 degrees celcius. The p-value is less than 2.2E-16 which is a value close to zero. If we test our model with 95% statistical significance, we can conclude that a number close to zero is less than 0.05 and say that our data is statistically significant. The residual value (r-squared) is 0.744, which means that 74.4% of variability in temperature can be explained by changes in depth.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
LTER_All_Regression <- lm(data = NTLLTER_processed, temperature_C ~ year4 +
                           daynum + depth)
summary(LTER_All_Regression)

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = NTLLTER_processed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.3086 -2.7609 -0.3194  2.5294  8.1964
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 123.85217   81.96700   1.511    0.132
## year4       -0.05591    0.04067  -1.375    0.170
## daynum        0.05268    0.00707   7.451 6.02e-13 ***
```

```
## depth          -1.92045      0.05271 -36.435 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.391 on 390 degrees of freedom
## Multiple R-squared:  0.7802, Adjusted R-squared:  0.7785
## F-statistic: 461.5 on 3 and 390 DF,  p-value: < 2.2e-16
```

```
step(LTER_All_Regression)
```

```
## Start:  AIC=966.18
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS    AIC
## - year4    1      21.7  4505.8  966.08
## <none>                 4484.0  966.18
## - daynum    1     638.3  5122.3 1016.62
## - depth     1    15263.4 19747.5 1548.29
##
## Step:  AIC=966.08
## temperature_C ~ daynum + depth
##
##           Df Sum of Sq    RSS    AIC
## <none>                 4505.8  966.08
## - daynum    1       717  5222.8 1022.27
## - depth     1    15242  19747.5 1546.29
##
##
## Call:
## lm(formula = temperature_C ~ daynum + depth, data = NTLLTER_processed)
##
## Coefficients:
## (Intercept)      daynum        depth
##    11.19860     0.05466    -1.91767
```

```
#10
LTER_Best_Regression <- lm(formula = temperature_C ~ daynum + depth, data = NTLLTER_processed)
summary(LTER_Best_Regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ daynum + depth, data = NTLLTER_processed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.6397 -2.6632 -0.3101  2.5174  8.0771
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11.19859    1.341363   8.349 1.21e-15 ***
## daynum       0.054660    0.006929   7.888 3.10e-14 ***
## depth       -1.917670    0.052729  -36.368 < 2e-16 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.395 on 391 degrees of freedom
## Multiple R-squared:  0.7792, Adjusted R-squared:  0.778
## F-statistic: 689.8 on 2 and 391 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC method recommend that we only use two explanatory variables, daynum and depth, to predict temperature in our multiple regression. The R-squared value is 0.7792 which means that both daynum and depth explain 77.9% of the observed variance in temperature. Yes, the R-squared model has improved from 74.4% (only depth) to 77.9% (both depth and daynum) for explanatory variable(s) to predict the observed variance in temperature.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12
Temp_anova <- aov(data = NTLLTER_processed, temperature_C ~ lakename)
summary(Temp_anova)

##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      4    228    56.96   1.098  0.357
## Residuals    389   20176    51.87

Temp_regression <- lm(data = NTLLTER_processed, temperature_C ~ lakename)
summary(Temp_regression)

##
## Call:
## lm(formula = temperature_C ~ lakename, data = NTLLTER_processed)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.331  -6.756  -2.550   7.338  14.536
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    10.5556     1.6975   6.218  1.3e-09 ***
## lakenamePaul Lake     1.9008     1.7856   1.065   0.288
## lakenamePeter Lake     2.0088     1.7904   1.122   0.263
## lakenameTuesday Lake  -0.4389     2.4006  -0.183   0.855
```

```
## lakenameward Lake      3.3755      2.1610      1.562      0.119
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.202 on 389 degrees of freedom
## Multiple R-squared:  0.01117,    Adjusted R-squared:  0.0009981
## F-statistic: 1.098 on 4 and 389 DF,  p-value: 0.3571
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: No, there is no significant difference because the p-value equals 0.357 which is greater than 0.05 meaning that there is no statistically significant (95%) difference between lakes and their temperatures in July.

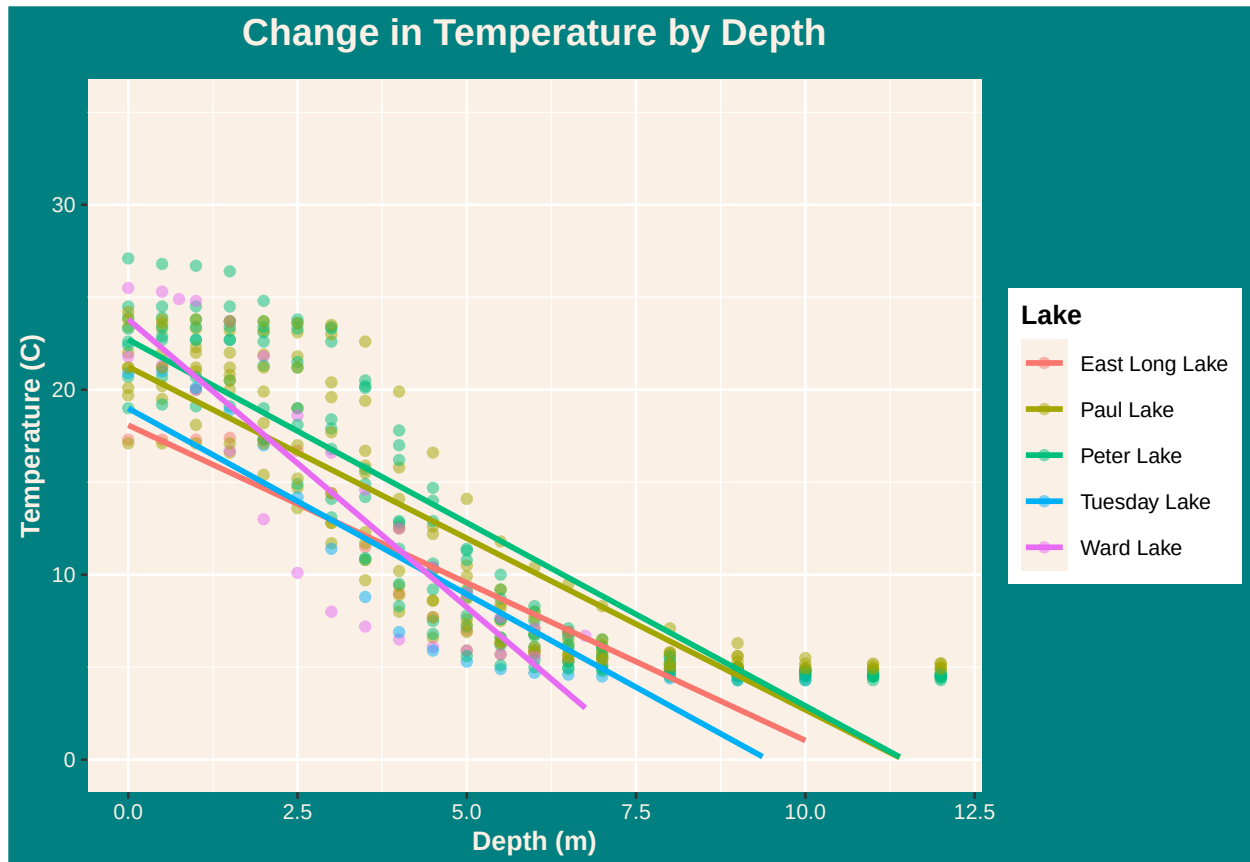
14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
Lake.Temp.depth <- ggplot(NTLLTER_processed, aes(x=depth, y=temperature_C,
                                                  color = lakenameward)) +
  ylim(0, 35) +
  geom_point(alpha = 0.5) +
  geom_smooth(method="lm", se = FALSE) +
  labs (title = "Change in Temperature by Depth",
        x = "Depth (m)",
        y = "Temperature (C)",
        color = "Lake")

print(Lake.Temp.depth)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 13 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```

15. Use the Tukey's HSD test to determine which lakes have different means.

```
#15
Lake.Temp.depth.groups <- HSD.test(Temp_anova, 'lakename', group = TRUE)
Lake.Temp.depth.groups$groups
```

```
##          temperature_C groups
## Ward Lake          13.93103    a
## Peter Lake          12.56438    a
## Paul Lake           12.45633    a
## East Long Lake      10.55556    a
## Tuesday Lake        10.11667    a
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: All lakes have the same mean temperature as Peter Lake. This means that none lakes are statistically distinct from all the other lakes (labeled as “a”) and therefore show no significant difference in temperature among different types of lakes.

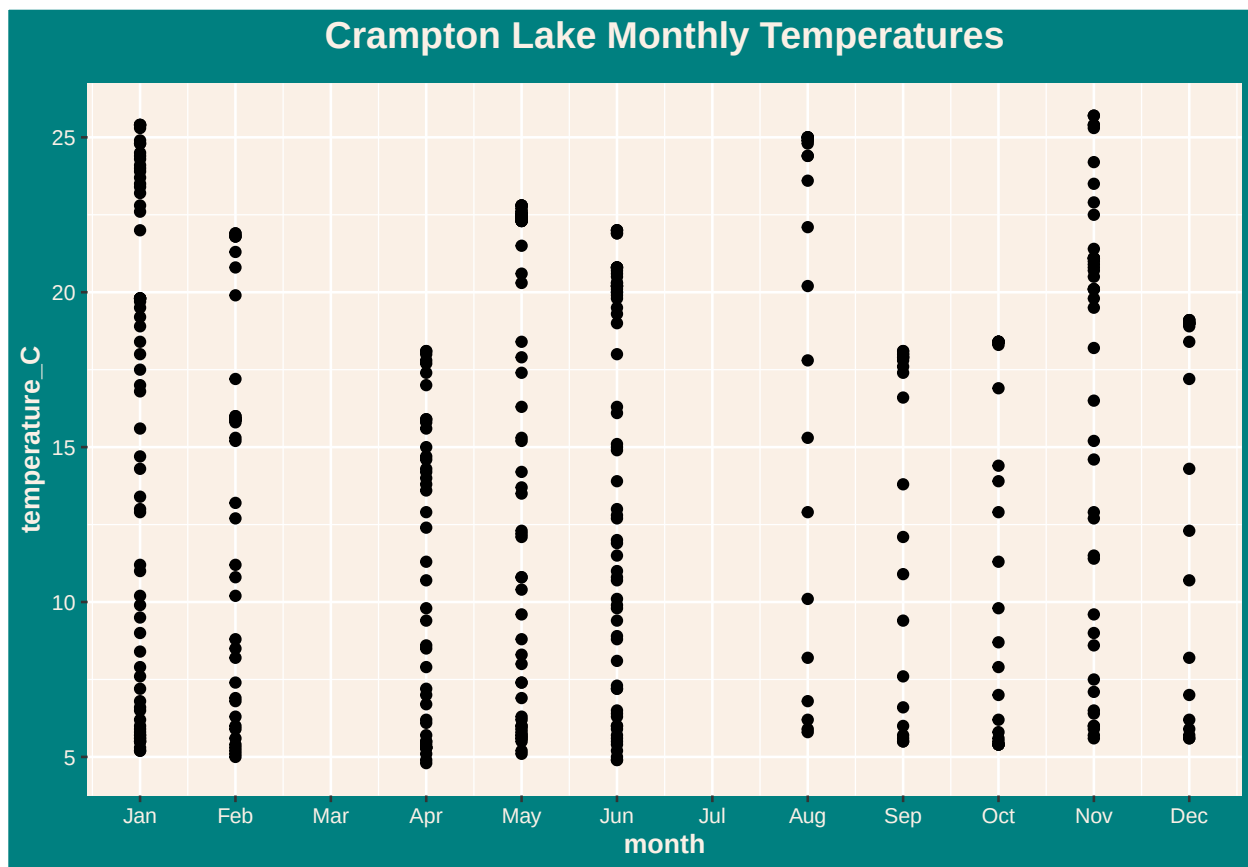
17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: Two-sample T-test will allow us to test the hypothesis that the mean temperature of two samples (Peter Lake and Paul Lake) are equivalent or not.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
Crampton.Temp <- NTLLTER_raw %>%
  mutate(month = month(sampledate)) %>%
  filter(lakename == "Crampton Lake") %>%
  select(lakename, month, year4, daynum, depth, temperature_C) %>%
  drop_na()

Crampton.plot <- ggplot(Crampton.Temp, aes(x = month, y = temperature_C)) +
  geom_point() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(title = "Crampton Lake Monthly Temperatures")
print(Crampton.plot)
```

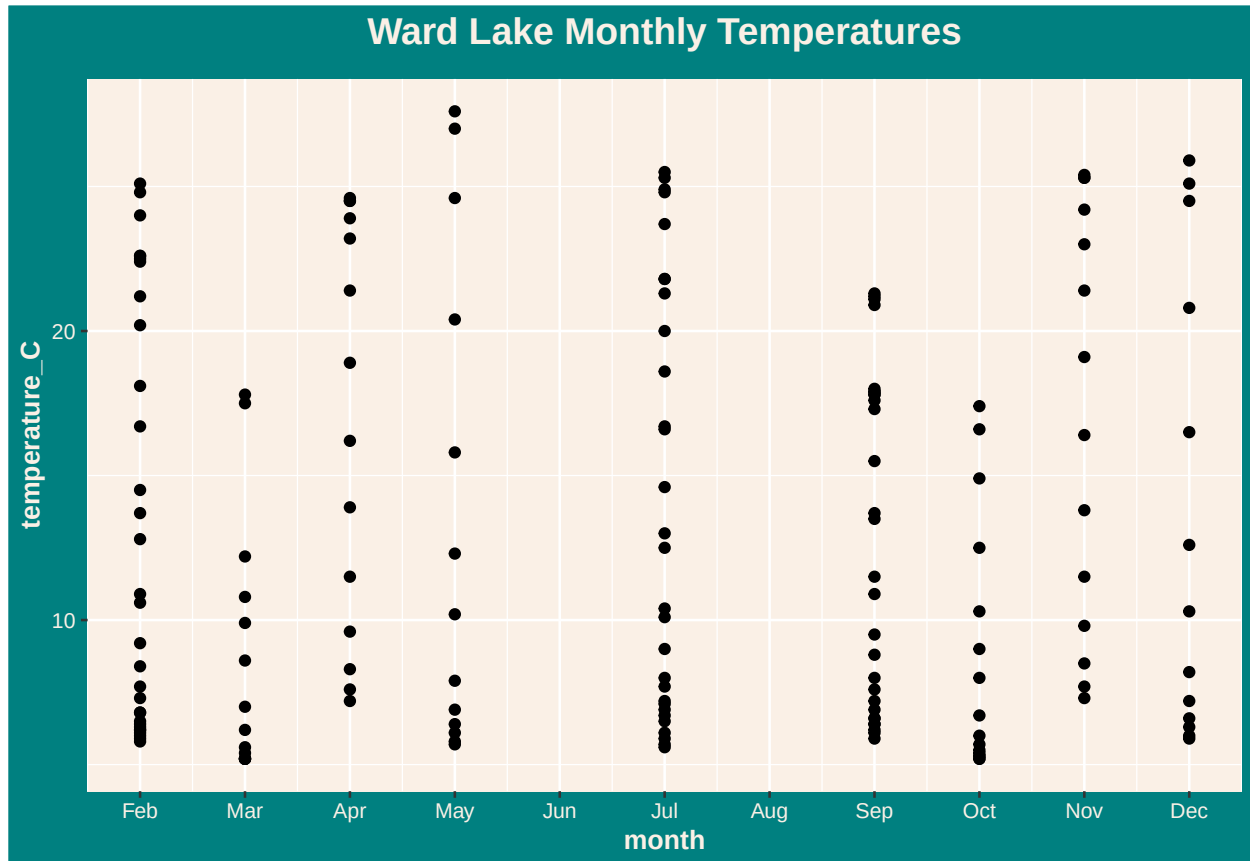


```
#There is no July data for Crampton Lake if we plot from NTLLTER_raw.
#Crampton Lake was not listed after wrangling data for NTLLTER_processed
#for only July data.
```

```
Ward.Temp <- NTLLTER_raw %>%
  mutate(month = month(sampledate)) %>%
  filter(lakename == "Ward Lake") %>%
  select(lakename, month, year4, daynum, depth, temperature_C) %>%
```

```
drop_na()

Ward.plot <- ggplot(Ward.Temp, aes(x = month, y = temperature_C)) +
  geom_point() + scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(title = "Ward Lake Monthly Temperatures")
print(Ward.plot)
```



```
##Note: I will try to wrangle for May temperature data for Crampton lake and
## Ward Lake, since they both have values within that month.
```

```
twosample.May.Temp <- NTLLTER_raw %>%
  mutate(month = month(sampledate)) %>%
  filter(month == "5") %>%
  select(lakename, month, year4, daynum, depth, temperature_C) %>%
  drop_na()

twosample.Lake.May <- twosample.May.Temp %>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

t_test <- t.test(data = twosample.Lake.May, temperature_C ~ lakename)
t_test
```

```
##
## Welch Two Sample t-test
##
```

```
## data: temperature_C by lakename
## t = 0.73848, df = 15.511, p-value = 0.4712
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to zero
## 95 percent confidence interval:
## -3.494617 7.216252
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.45312                13.59231
```

```
t_test_result <- lm(data = twosample.Lake.May, temperature_C ~ lakename)
summary(t_test_result)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = twosample.Lake.May)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3531  -7.4531  -0.1531   6.9469  14.0077
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      15.4531     0.9122  16.940  <2e-16 ***
## lakenameWard Lake  -1.8608     2.2201  -0.838   0.405
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.298 on 75 degrees of freedom
## Multiple R-squared:  0.00928,    Adjusted R-squared:  -0.003929
## F-statistic: 0.7025 on 1 and 75 DF,  p-value: 0.4046
```

Answer: If we compare the mean temperature for Crampton Lake and Ward Lake in May, we found that there was a p-value of 0.4712 which is greater than 0.05. This means that we fail to reject the null hypothesis that mean temperatures for Crampton Lake and Ward Lake are equal in May. We expect for the mean May temperatures to be different for the two lakes. The t-test estimates that the mean May temperature estimates are greater in Crampton Lake (15.45312 degrees C) than Ward Lake (13.59231 degrees C). The 95% confidence interval for the mean difference is about -3.49 to 7.22 which represents that there can be a difference in values or possibly no difference if the difference is zero.